

Christopher Z. Cui

Ph.D. Student, Computer Science

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EDUCATION

University of California, San Diego

September 2024–Expected: June, 2030

PhD in Computer Science and Engineering; Advisor: **Dr. Prithviraj Ammanabrolu**

Georgia Institute of Technology

August 2021–May 2024

M.S. in **Interactive Intelligence**; GPA: 4.00/4.00; Advisors: **Dr. Mark O. Riedl**, **Dr. Thad Starner**

University of North Carolina, Chapel Hill

August 2016–May 2020

B.S. in Computer Science; GPA: 3.72/4.00

RESEARCH EXPERIENCE

Research Intern

Incoming: June 2026–August 2026

Microsoft Research, under Marc-Alexandre Côté & Xingdi “Eric” Yuan

Montréal, Québec

- Incoming summer research internship continuing collaboration on LLM agents and situated text environments.

Graduate Research Assistant

September 2024–Current

University of California, San Diego, under Dr. Prithviraj Ammanabrolu

San Diego, California

- **Behavior Cue Reasoning: Monitorable Reasoning Improves Efficiency and Safety through Oversight** (Christopher Z. Cui, Taylor W. Killian, Prithviraj Ammanabrolu; pre-print): designed and implemented a training method that makes LLM reasoning easier to monitor and control, improving both efficiency and safety across math and agent benchmarks.
 - Reduced unnecessary reasoning tokens by up to **50%**.
 - Improved safe task success from **46%** to **96%** by detecting and redirecting unsafe reasoning paths.
 - Built training and evaluation pipelines for **Qwen3-8B**, **GLM-Z1-9B**, **AIME**, **TextWorld**, and **HazardWorld**.
- **TALES: Text Adventure Learning Environment Suite** (Christopher Z. Cui, Xingdi Yuan, Ziang Xiao, Prithviraj Ammanabrolu, Marc-Alexandre Côté; **Oral, REALM 2025**; non-archival): helped build and analyze a benchmark suite for evaluating LLM agents in interactive text environments, in collaboration with Microsoft Research.
 - Benchmarked **57 model configurations** across **122 tasks**.
 - Led paper writing, construction of automated agent failure detectors, and analysis of model behavior traces.
 - Presented as an **oral presentation at REALM 2025**.
- Lead system installation and configuration for the lab's local GPU compute clusters in partnership with UCSD campus IT.

AI Engineer

June 2025–September 2025

MBZUAI Institute of Foundation Models, under Dr. Taylor W. Killian

Sunnyvale, California

- Investigated context management strategies for multi-turn RL training of **LLM agents**, specifically the effect of retaining past reasoning traces in-context and combining them with self-critique reasoning. Found **up to 20% in-distribution performance gains** on agentic text-environment benchmarks.
- Trained **Qwen3-8B** with **PPO** atop the **verl-agent** framework across four agentic text environments (TextWorld, TextWorld Express, ALFWorld, Lifegate) on H200 clusters.
- Built a custom stochastic text environment (Lifegate) for stress-testing risk-sensitive RL training of LLM agents.

Graduate Research Assistant

January 2023–July 2024

Georgia Tech, under Dr. Thad Starner

Atlanta, Georgia

- Helped secure funding by presenting progress on educational AI tools to faculty and College of Computing leadership.
- Research Lead for **Socratic Mind**, an AI-powered platform for scalable, interactive oral assessment.
 - Winner of the **2023-24 Tools Competition in Educational Technologies**
 - Garnered interest from 15+ professors across 7 universities
 - Successfully served over 1,000 college students in graduate and undergraduate level courses
- Led research teams building AI tools for plagiarism detection, online exam integrity, answer watermarking, and coding-assignment support.
- Led classroom AI research and **Impasse**, an IDE extension for detecting when students are struggling in coding assignments.

Graduate Research Assistant

March 2022–May 2024

Georgia Tech, under Dr. Mark O. Riedl

Atlanta, Georgia

- Built story-planning pipelines that combine language models with symbolic planning to generate coherent plot skeletons.
- Improved text-game environment tooling by streamlining map creation for user-generated layouts in LIGHT TextWorld.

- Developed an integration wrapper for training and evaluating reinforcement-learning agents in LIGHT TextWorld.
- Explored few-shot task transfer for open-ended text-game agents using prompt tuning and ensemble guidance.

PUBLICATIONS

Behavior Cue Reasoning: Monitorable Reasoning Improves Efficiency and Safety through Oversight

Christopher Z. Cui, Taylor W. Killian, Prithviraj Ammanabrolu

Pre-print

TALES: Text Adventure Learning Environment Suite

Christopher Z. Cui, Xingdi Yuan, Ziang Xiao, Prithviraj Ammanabrolu, Marc-Alexandre Côté

Oral Presentation, REALM 2025 (non-archival)

Leveraging Past Assignments to Determine If Students Are Using ChatGPT for Their Essays

Yuhui Zhao, Chunhao Zao, Rohit Sridhar, Christopher Cui, Thad Starner

Proceedings of the Tenth ACM Conference on Learning@ Scale 2024

Examinator v4.0: Cheating Detection in Online Take-Home Exams

Christopher Cui*, Jui-Tse Hung*, Vaibhav Malhotra, Hardik Goel, Raghav Apoorv, Thad Starner (*shared first author)

Proceedings of the Tenth ACM Conference on Learning@ Scale 2024

Socratic Mind: Scalable Oral Assessment Powered By AI

Jui-Tse Hung, Christopher Cui, Diana M. Popescu, Saurabh Chatterjee, Thad Starner

Proceedings of the Tenth ACM Conference on Learning@ Scale 2024

Answer Watermarking: Using Answer Generation Assistance Tools to Find Evidence of Cheating

Christopher Cui, Jui-Tse Hung, Pranav Sharma, Saurabh Chatterjee, Thad Starner

Proceedings of the Tenth ACM Conference on Learning@ Scale 2024

A Mixture-of-Experts Approach to Few-Shot Task Transfer in Open-Ended Text Worlds

Christopher Cui, Xiangyu Peng, Mark Riedl

Pre-print

Thespian: Multi-Character Text Role-Playing Game Agents

Christopher Cui, Xiangyu Peng, Mark O. Riedl

Proceedings of the AIIDE Workshop on Experimental AI in Games (EXAG) 2023

Story Shaping: Teaching Agents Human-like Behavior with Stories

Xiangyu Peng*, Christopher Cui*, Wei Zhou, Renee Jia, Mark O. Riedl (*shared first author)

Proceedings of the 19th AAAI Conference on Artificial Intelligence in Interactive and Digital Entertainment (AIIDE) 2023

Examinator v3.0: Cheating Detection in Online Take-Home Exams

Jui-Tse Hung, Christopher Cui, Varun Agarwal, Saurabh Chatterjee, Raghav Apoorv, Rocko Graziano, Thad Starner

Proceedings of the Tenth ACM Conference on Learning@ Scale 2023

Neural Story Planning

Anbang Ye, Christopher Cui, Taiwei Shi, Mark O. Riedl

Proceedings of the AAAI Workshop on Creative AI Across Modalities 2023

SERVICE & REVIEWING

- **Gold-tier Reviewer (top 25%)**, ICML 2026
- Reviewer, NeurIPS 2025
- Reviewer, ICLR 2026

SKILLS

- Technical: Large Language Models, Reinforcement Learning, RLHF / PPO / GRPO, Natural Language Processing, Distributed Training, LLM Reasoning, AI Agents
- Languages: Python, C/C++, Java, JavaScript
- Tools and Frameworks: PyTorch, Hugging Face Transformers, verl / verl-agent, OpenRLHF, vLLM, Ray, DeepSpeed